

Airlines Flights Analysis

1. Problem Statement

The airline industry handles large volumes of travel and pricing data daily. Airlines require analytical insights to understand passenger travel patterns, route performance, pricing behavior, and operational efficiency. In this project, students will act as Data Analysts and perform exploratory data analysis on airline travel data to identify trends, patterns, and business insights related to airline operations and fare distribution.

2. Data Dictionary

Column Name	Description
Airline	Name of the airline company
Date_of_Journey	Journey travel date
Source	Departure city/location
Destination	Arrival city/location
Route	Complete travel route
Dep_Time	Age of the individual involved in the incident.
Arrival_Time	Gender of the individual.
Duration	Ethnicity or racial background of the individual.
Total_Stops	City where the incident took place.
Additional_Info	State where the incident occurred.
Price	Ticket fare amount

3. Important KPIs / Measures

1. Total Flights
2. Average Ticket Price
3. Maximum Ticket Price
4. Minimum Ticket Price
5. Average Travel Duration
6. Total Airlines
7. Total Source Cities
8. Total Destination Cities
9. Most Popular Route
10. Highest Revenue Generating Airline

4. Storytelling Visualizations

1. Which airline generated the highest revenue during the analysis period?
2. How does ticket price vary across different cabin classes
3. What are the peak flight booking/travel hours and how does flight activity change throughout the day?
4. How does the number of stops impact the average ticket price of flights?
5. Which flight routes are most frequently used and what is their average travel duration?